

Framestore FUSE - An Unreal Engine Pipeline for Large Scale Live-Action VFX Production

Nestor Prado
Framestore
London, UK
nestor.prado@framestore.com

Theo Jones
Framestore
London, UK
theo.jones@framestore.com

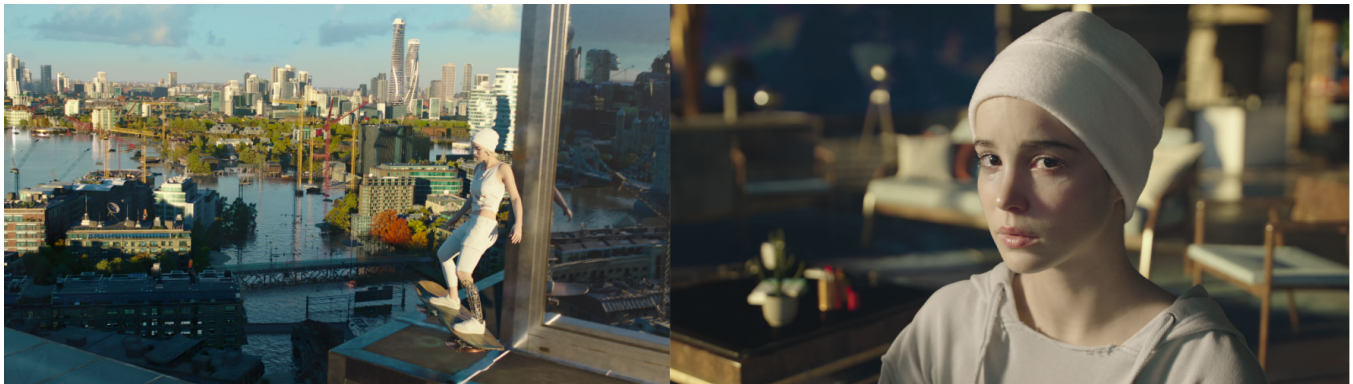


Figure 1: FLITE, 2023.

ABSTRACT

Framestore has a robust and mature visual effects pipeline that has been proven across numerous complex, large-scale productions. Additionally, Framestore has a history of producing innovative smaller-scale realtime projects in the areas of VR and experiential entertainment. In this talk, we present FUSE, the Framestore Unreal Shot Engine, a scaleable toolset and pipeline that allows for the power of a realtime engine, Unreal, and its capacity for interactivity and rapid iterations to be utilised as the hub and final renderer on a complex, multi-disciplinary, live-action, visual effects production. This entirely novel pipeline has been used to create FLITE, a 15 minute live-action, CG hybrid movie with realtime workflows at its core. This approach has empowered traditional visual effects artists within a realtime workflow and has also transformed the film's creative process.

CCS CONCEPTS

• Computing methodologies → Computer graphics.

KEYWORDS

VFX, real-time, Unreal Engine

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1 BACKGROUND

Framestore has been using Unreal Engine and realtime software within its Immersive and Virtual Production divisions for well over a decade. Historically the scale of projects within these divisions was always restricted and therefore the need for a large-scale, multi-departmental, custom toolsets was limited. Conversely over this period the Framestore Film & Episodic division has continued to develop a mature and sophisticated pipeline for delivering large-scale, complex productions involving hundreds, or even thousands, of shots. With the increased capabilities of realtime engines such as Unreal it was clear that the gap between these approaches needed to be bridged.

2 PRODUCTION REQUIREMENTS

Modern, large-scale, visual effects pipelines are designed to facilitate extensive teams of specialist artists working on a large number of highly complex shots while maintaining a consistently high quality of visual output and the ability to address precise creative notes. As such they tend to favour robust repeatability and granular creative control. Iteration speed can be affected as well as artist inter-departmental collaboration and the ability to view their work in its wider context. Realtime engines on the other hand tend to excel at rapid iteration cycles providing a more holistic view of a shared project and scope for interactive collaboration. However they are not designed to work with the full range of VFX departmental

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disciplines, to fit into a live action compositing workflow or to deliver consistent repeated results. The primary challenge faced in the design of FUSE was trying to bridge the gap between the strength of both approaches. As such a number of key requirements were formulated:

- Integrate seamlessly into existing VFX pipeline tools and review processes where appropriate
- Enable traditional VFX artists from multiple departments to be productive without the need for extensive retraining
- Provide concurrent asset version access in order to facilitate instant comparison of asset/shot versions
- Enable multipass rendering in a manner that facilitates visual effects compositors integrating CG material with live action footage
- Enable robust, repeatable rendering with ability to trace back asset versions and creative approvals of content

3 IMPLEMENTATION

3.1 Integration into Existing VFX Toolsets

Framestore has an extensive set of publishers that facilitate artists to publish and quality control their work in a fully trackable manner. For FUSE we adapted all these publishers to work in Windows and output data compatible with the needs of Unreal Engine. This data is automatically synced in to Unreal via Perforce utilizing our TeamCity infrastructure. Correlation between assets in Unreal and their counterparts in the Framestore VFX Asset Database is always maintained. The creation of managed versions of these assets in Unreal is also automated, making the process of publishing an asset and being able to utilize it in Unreal transparent for the artists. This whole process is customizable to fit any number of pipelines that aren't Framestore-specific. As an example, this automatic process allowed animation done outside Unreal to be rendered through Unreal without user intervention greatly improving the creative feedback loop.

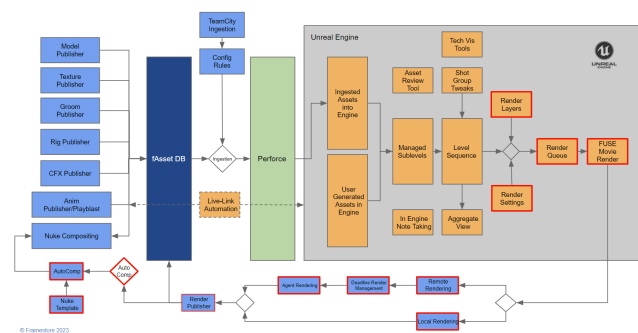


Figure 2: FUSE - Pipeline/Toolset, high level view

3.2 Asset Management

At the core of FUSE is a newly implemented asset management system (MAS) designed to augment Unreal Engine's content management systems. It is designed to make large-scale, version-driven linear content projects easier to work with. Key features are:

- Concurrent versions of assets/shots can be accessed locally allowing rapid iteration
- Related versions can be conveniently swapped out depending on the context providing flexibility of having high end VFX assets as well as realtime representations
- Compatible with unmodified versions of Unreal Engine for sharing with vendors that don't have access to this system

3.3 Shot Management

The asset management system extended to shot related assets as well. A key difference between traditional Unreal Engine shot workflows and FUSE is that the shot entity (represented in Unreal by a level sequence) contains a direct correlation to the sub-levels that are needed to dynamically "build" each shot.

Another key aspect of our shot workflows is the introduction of Shot Group Tweaks. These allow the storage and deployment of modifications done to any actor on a shot to any number of other shots.

3.4 Rendering

Even-though Unreal Engine ships with multi-layering capabilities and a render queue manager/dispatcher there was a need to augment and modify some of these capabilities to allow for scalable workflows involving rendering dozens of render layers per shot. Key features of this custom rendering system are:

- Decoupling of render layers and sub-levels
- Rule-based render layer assignments
- Managed render layers
- Custom editor UI to visualise/author/check render layers

4 CONCLUSIONS

FUSE has been successfully used in the production of a highly complex 15min live-action, CG hybrid Film. FLITE. This project has demonstrated the power of correctly utilising realtime engines as the hub of a live-action, linear narrative, content creation pipeline. The increased iteration speeds and interactive collaboration improved efficiency but also created new creative opportunities allowing parallel workflows and creative flexibility throughout the process. Although more work is required in certain key areas including rendering, the signs are clear that they can play an increasingly important role in the future of VFX production.

ACKNOWLEDGMENTS

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